Summary in Graph

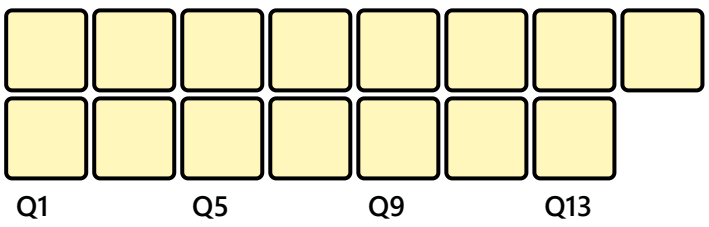
Exam Summary (GO Classes CS | CO and Architecture | Cache Memory | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (28) out of 281 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 CO & Architecture

Which of the following mapping is not used for mapping process in cache memory

A. Associative mapping

B. Direct mapping

C. Set-Associative mapping

D. Segmented – page mapping

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

The principle of Locality of reference justifies the use of:

A. Virtual memory

B. Interrupts

C. Cache memory

D. Secondary memory

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

More than one word is put in one cache block to:

A. exploit the temporal locality of reference in a program

B. exploit the spatial locality of reference in a program

C. reduce the miss penalty

D. none of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

The tag-search used in a fully associative cache memory is

A. Parallel search

B. Sequential search

C. Binary search

D. Selection search

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

In a cache memory, if the total number of sets is 's', then the set offset is:

A. 2^8

B. $\log_2 s$

C. s^2

D. s

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A direct-mapped cache with 128 KB of data is designed using 32-bit addresses. The cache uses a block size of 4 bytes. Each cache line has: A valid bit (1 bit), A tag field, data block. What is the total size of the cache (in bits), including both the data array and the tag array?

A. 2 Mbits

B. 1.7 Mbits

C. 2.5 Mbits

D. 1.5 Mbits

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a machine with a byte-addressable main memory of 2^{16} bytes and a block size of 8 bytes. Assume that a direct-mapped cache consisting of 32 lines is used with this machine. How many bits will be there in Tag, line and word fields of the format of main memory addresses?

A. $8, 5, 3$

B. $8, 6, 2$

C. $7, 5, 4$

D. $7, 6, 3$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a small 2-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently used (LRU) scheme. The number of cache misses for the following sequence of block addresses is $8, 12, 0, 12, 8$.

A. 2

B. 3

C. 4

D. 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A cache memory needs an access time of 30 ns and main memory 150 ns, what is the average access time of the CPU (assume hit ratio $= 80\%$)?

A. 60 ns

B. 30 ns

C. 150 ns

D. 70 ns

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A two-way set associative cache memory unit with a capacity of $16\,KB$ is built using a block size of $8\,words$. The word length is $32 - bits$. The physical address space is $4\,GB$.
The number of bits in the **TAG, SET** fields are

A. $20, 7$

B. $19, 8$

C. $20, 8$

D. $21, 9$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

In a particular system it is observed that the cache performance gets improved as a result of increasing the block size of the cache. The primary reason behind this is:

A. Programs exhibits temporal locality

B. Programs have small working set

C. Read operation is frequently required rather than write operation

D. Programs exhibits spatial locality

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A direct-mapped cache is of size 32 KB and has a block size of 32 Bytes. The CPU also generates a 32-bit address. Number of bits needed for indexing the cache:

A. 14

B. 15

C. 10

D. 17

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A computer system uses $16-$bit memory addresses. It has a $16KB$ cache organised in a $4-$way set associative manner with 64 bytes per cache block. Assume that the size of each memory word is 2B. When a program is executed ,The processor reads data sequentially from the following byte addresses:

$$128, 144, 2176, 2180, 128, 2176$$

All the above addresses are shown in decimal values. Assume the cache is initially empty. The number of addresses hit in the cache?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a computer with a 12-bit address space and a byte-addressable memory. It has a data cache capable of holding eight cache blocks. Each cache block is 4 bytes. During a particular program execution, the processor issues memory read instructions to the following sequence of block addresses:

$$128, 129, 130, 131, 189, 188, 257, 256, 128, 129, 131, 189, 130, \text{ and } 188.$$

If a 2-way set-associative cache that uses the LRU-replacement algorithm and the cache is initially empty. The number of blocks that are hit in the cache of program is _____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is _____

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 02sec | Discuss

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